

Hwk class 12

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Q13 Read this file into R and determine the sample size for each genotype and their corresponding median expression levels for each of these genotypes.

Hint: The `read.table()`, `summary()` and `boxplot()` functions will likely be useful here. There is an example R script online to be used ONLY if you are struggling in vein. Note that you can find the median value from saving the output of the `boxplot()` function to an R object and examining this object. There is also the `median()` and `summary()` function that you can use to check your understanding.

```
url <- "https://bioboot.github.io/bggn213_W19/class-material/rs8067378_ENSG00000172057.6.txt"
data <- read.table(url, header = TRUE)
head(data)
```

	sample	geno	exp
1	HG00367	A/G	28.96038
2	NA20768	A/G	20.24449
3	HG00361	A/A	31.32628
4	HG00135	A/A	34.11169
5	NA18870	G/G	18.25141
6	NA11993	A/A	32.89721

```
table(data$geno)
```

A/A	A/G	G/G
108	233	121

```
summary(data$exp)
```

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
6.675	20.004	25.116	25.640	30.779	51.518

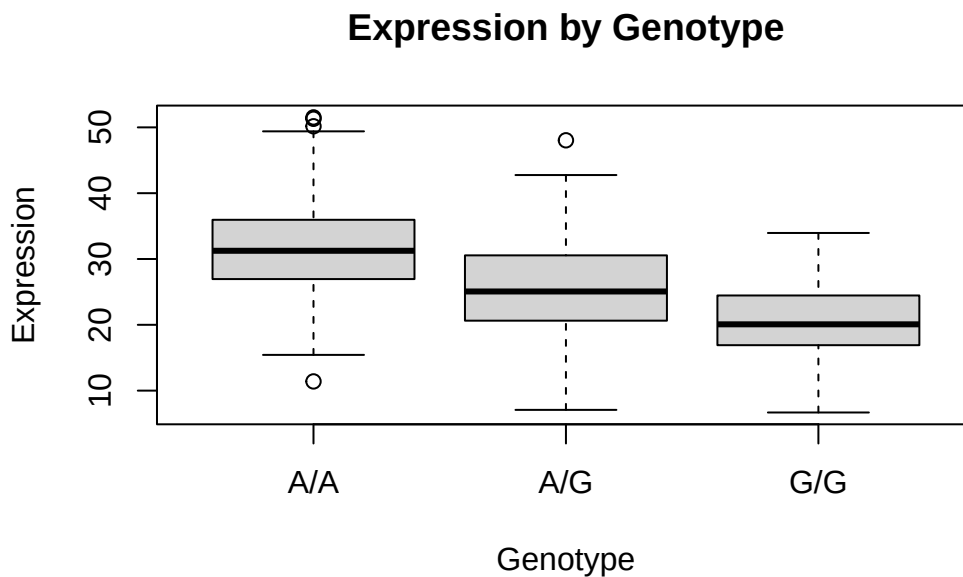
```
tapply(data$exp, data$geno, median)
```

A/A	A/G	G/G
31.24847	25.06486	20.07363

The file was read into R the sample size for each genotype is 108 for A/A, 233 for A/G, 121 for G/G. The median expression levels was 31.24847 for A/A, 25.06486 for A/G, and 20.07363 for G/G.

Q14: Generate a boxplot with a box per genotype, what could you infer from the relative expression value between A/A and G/G displayed in this plot? Does the SNP effect the expression of ORM DL3? Hint: An example boxplot is provided overleaf – yours does not need to be as polished as this one.

```
boxplot(data$exp ~ data$geno, main= "Expression by Genotype", xlab= "Genotype", ylab="Expression")
```



The boxplot demonstrates that the A/A genotype has the highest expression value than A/G and G/G. This shows that the SNP does affect the expression of ORM DL3 as there are varying expression levels for each genotype. The G allele may be associated with lower expression based on the plot, more analysis would need to be conducted to fully conclude this statement.